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Writer Identification and Writer Retrieval

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Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg

WiSe 25/26



Motivation



Source: Nitramica Arts (CC-BY-SA 2.0), Max Pixel (CC-0)

Motivation

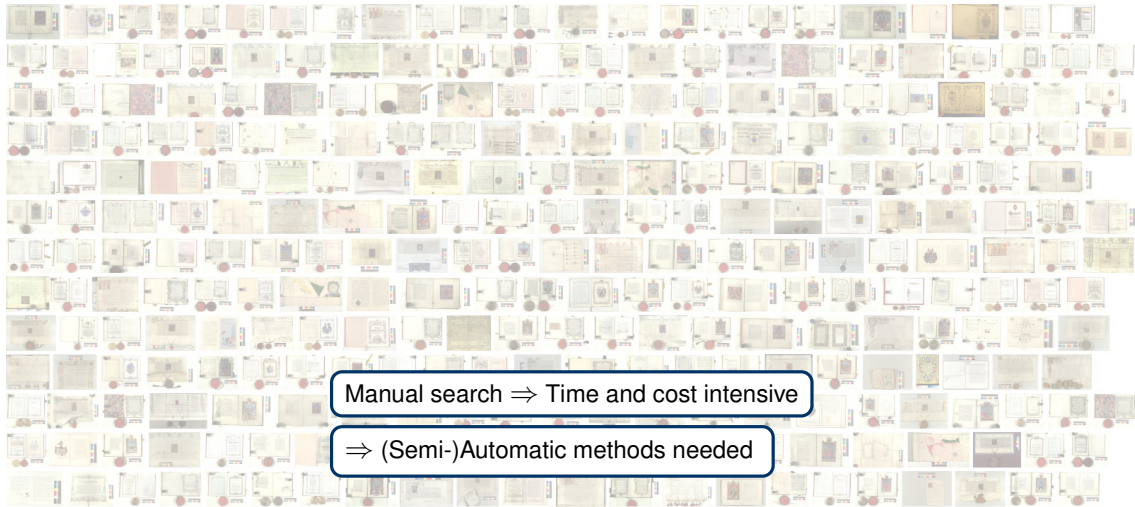


Source: Nitramica Arts (CC-BY-SA 2.0), Davide Iliff (CC-BY-SA 3.0)

Handwriting Analysis



Handwriting Analysis



Manual search $\Rightarrow$ Time and cost intensive

$\Rightarrow$ (Semi-)Automatic methods needed

Outline

Introduction to Writer Identification/Retrieval

General Approach

Sum Pooling vs. Generalized Max Pooling

Exemplar Classification

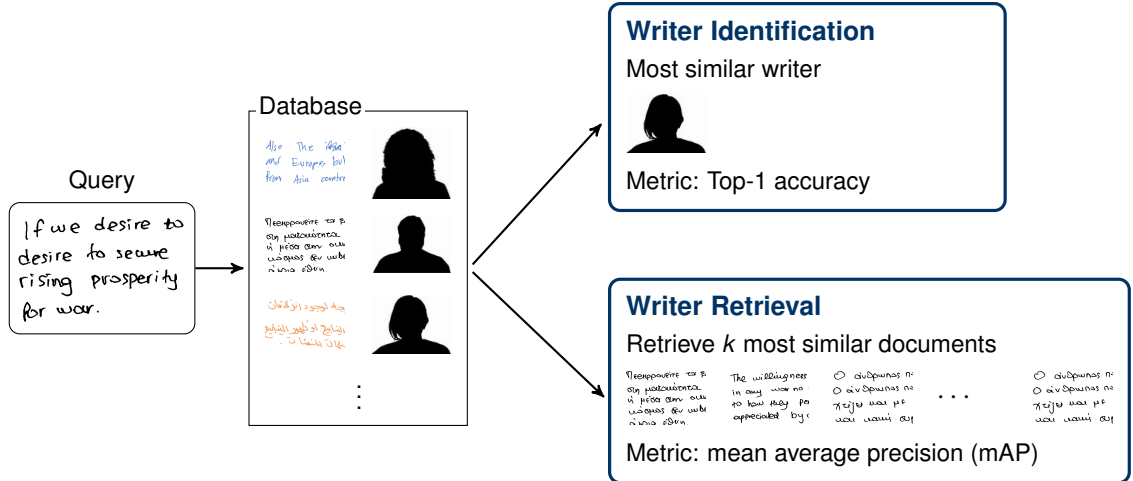


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Introduction to Writer Identification/Retrieval



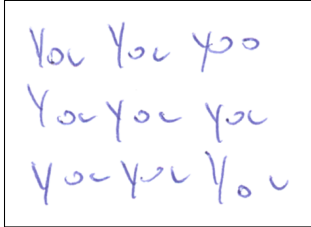
Writer Identification vs. Writer Retrieval



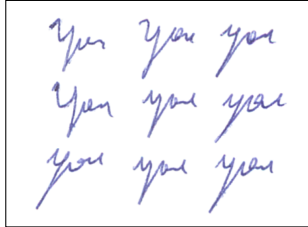
Source: ICDAR13 dataset, QUWI15 dataset, freepik.com

Challenges: Internal Factors

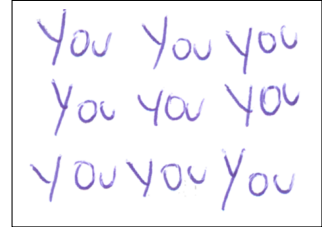
Writer A



Writer B



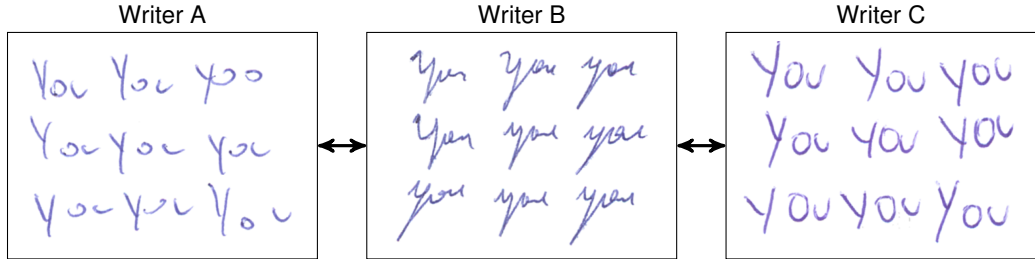
Writer C



- Within-writer variability

Source: CVL dataset [1], img-ids (left to right): 0001-7, 0022-7, 0021-7

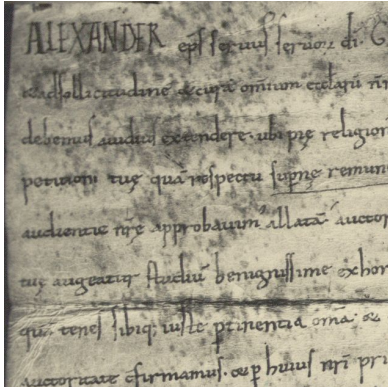
Challenges: Internal Factors



- Within-writer variability
- Between-writer variability

Source: CVL dataset [1], img-ids (left to right): 0001-7, 0022-7, 0021-7

Challenges: External Factors



Source: Göttingen Academy of Sciences and Humanities, JL 4490, 4671.

- Pen
- Document Material
- Artifacts

Contemporary Datasets

The willingness with which
in any war no matter how
to how they perceive veterans
appreciated by our nation.

Περαιτέρω τα βιβλία εορτίζουν
στη παρουσίαση των προτύπων
η μέρα των ομεινών. Αλλά ο
νόμος δεν υποβιβάζεται από

ICDAR13 benchmark dataset¹

- 4 documents per writer (2 English, 2 Greek)
- Train: 100 writers → 400 samples
- Test: 250 writers → 1000 samples

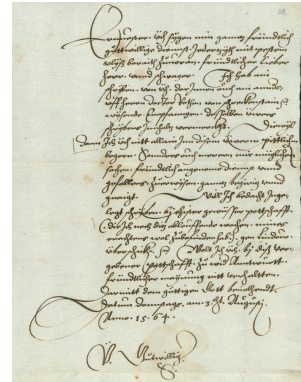
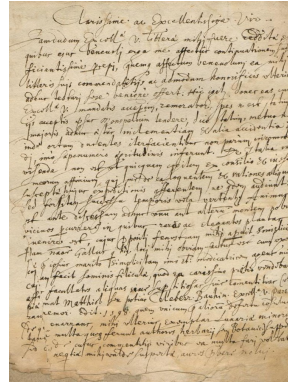
Other datasets: CVL (English, German), KHATT (Arabic), IAM (English)

¹G. Louloudis, B. Gatos, N. Stamatopoulos, and A. Papandreou, "ICDAR 2013 Competition on Writer Identification," in *ICDAR*, Washington DC, NY, Aug. 2013, pp. 1397–1401.

Historical Dataset

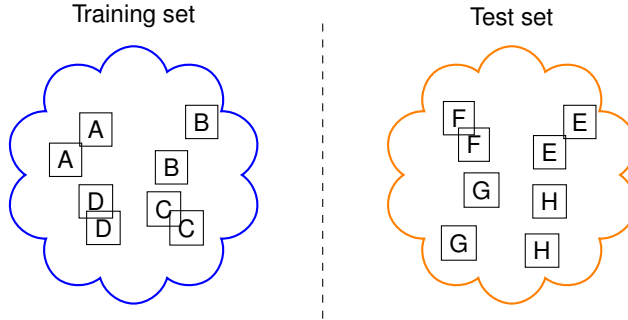
ICDAR17 competition dataset

- Letter collection (University library Basel)
- Train: 394 writers x 3 images
→ 1182 images
- Test: 720 writers x 5 images
→ 3600 images



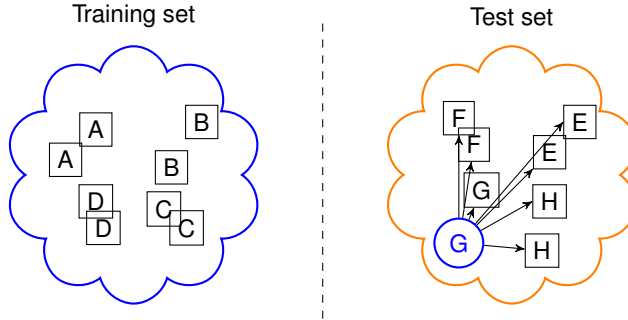
Source: ICDAR17 Historical-WI, ID: 2056-IMG_MAX_320331, 1146-3-IMG_MAX_1207684

Writer-Independent Datasets



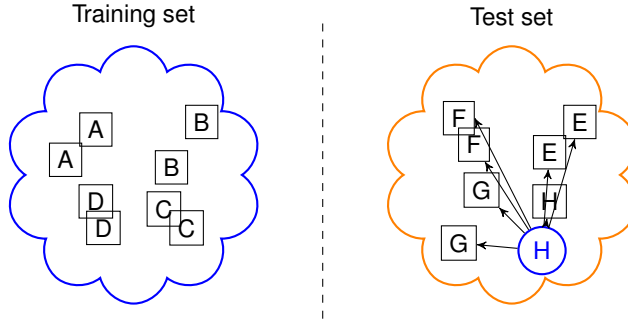
Training and test sets are independent
⇒ No training for a specific writer possible!

Writer-Independent Datasets



Training and test sets are independent
⇒ No training for a specific writer possible!

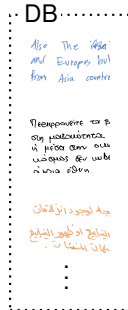
Writer-Independent Datasets



Training and test sets are independent
 $\Rightarrow$ No training for a specific writer possible!

Error Metrics

If we desire to
desire to secure
rising prosperity
for war.



Rank

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Identification rate

Mean precision at rank 1

Mean average precision

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad N: \text{\#queries}$$

Error Metrics

If we desire to
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DB

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and Europe but
from Asia countries

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Rank

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1

0

Identification rate

Mean precision at rank 1

Mean average precision

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad N: \text{\#queries}$$

$$AP_i = \frac{\sum_{k=1}^Q \Pr(k) \cdot \text{rel}(k)}{\text{number of relevant documents}}$$

Error Metrics

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25.3 million

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rel(k)	0	1	1	0
Pr(k)	0	0.5	0.6	0

Identification rate

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Error Metrics

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Rank

k 1 2 3 Q

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North Amen
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25.3 million

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rel(k)	0	1	1	0
Pr(k)	0	0.5	0.6	0

$$AP = (0.5 + 0.6)/2 \approx 0.58$$

Identification rate

Mean precision at rank 1

Mean average precision

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad N: \text{\#queries}$$

$$AP_i = \frac{\sum_{k=1}^Q Pr(k) \cdot rel(k)}{\text{number of relevant documents}}$$

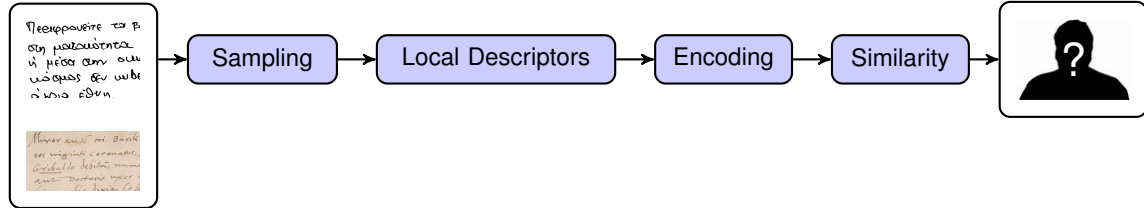


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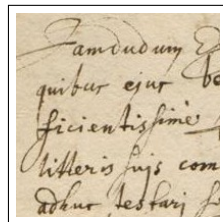
General Approach



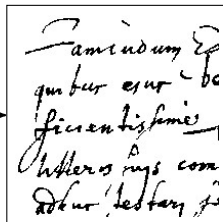
Methodology



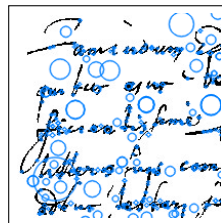
Sampling



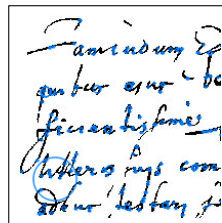
Binary image



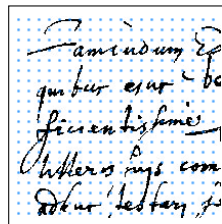
SIFT keypoints



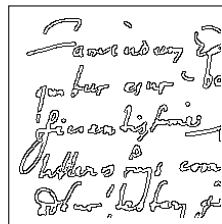
Restricted SIFT keypoints



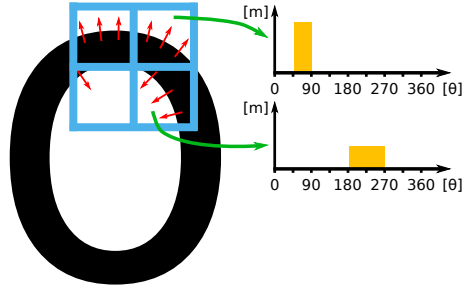
Dense



Contours



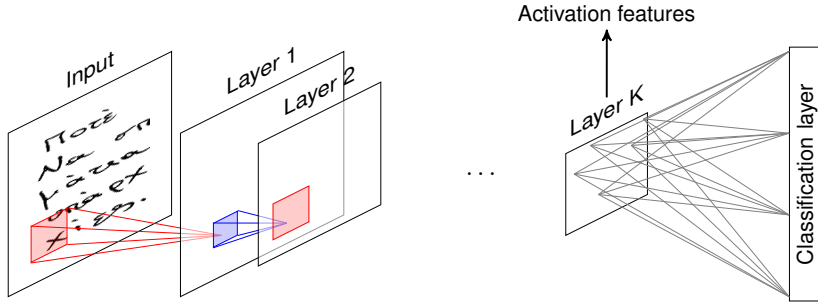
Handcrafted Features



SIFT: Scale-Invariant Feature Transform²

²D. G. Lowe, "Distinctive Image Features from Scale-Invariant Keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, Nov. 2004

Convolutional Neural Network Activation Features (CNN AF)



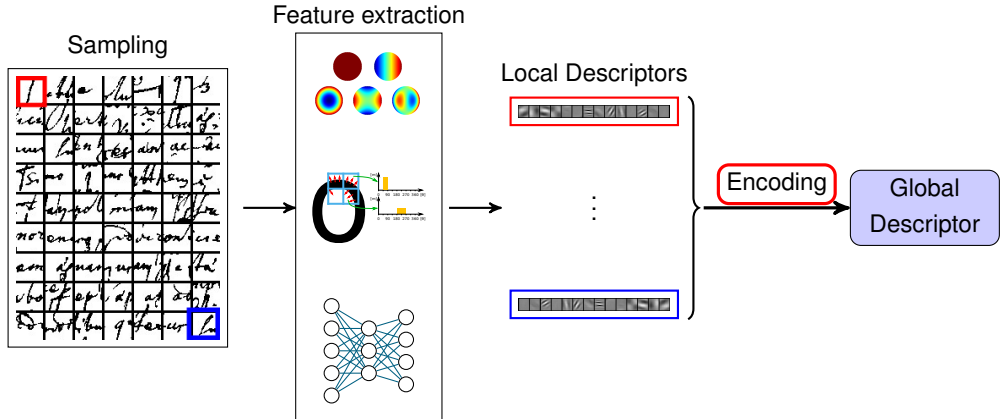
- Recall: no end-to-end training possible $\Rightarrow$ one-shot learning
- $\Rightarrow$ Surrogate task: classify writers of the training set using cross-entropy (“soft-max loss”)
- Use CNN as feature extractor

Metric Learning-based Features

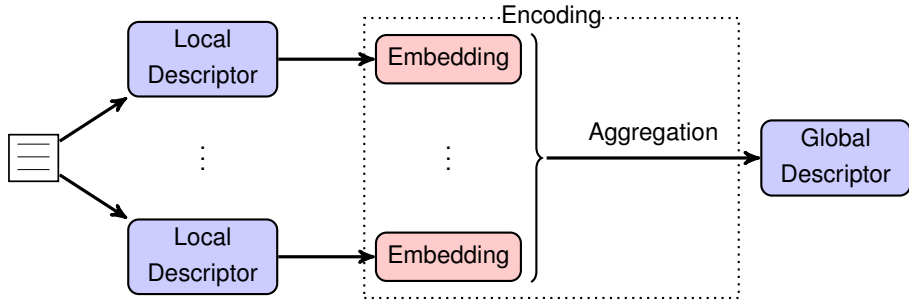
- Contrastive Loss
- Triplet Loss
- Magnet Loss
- Histogram Loss
- . . .

Unsupervised: e.g. AutoEncoders

Global Representation



Encoding

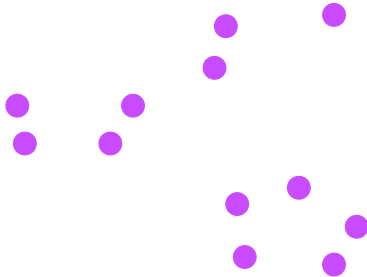


- Embedding: Map into high dimensional representation
- Aggregation: Sum pooling, generalized max-pooling³
- Normalization + Decorrelation

³N. Murray, H. Jegou, F. Perronnin, and A. Zisserman, "Interferences in Match Kernels," *TPAMI*, vol. 39, no. 9, 2016.

VLAD Embedding

VLAD: Vectors of Locally Aggregated Descriptors⁴

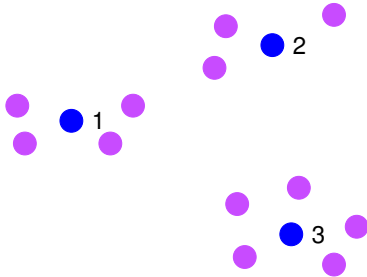


Local Descriptors: $\mathcal{X} = \{\mathbf{x}_i \in \mathbb{R}^D, i = 1, \dots, T\}$

⁴H. Jégou, F. Perronnin, M. Douze, J. Sánchez, P. Pérez, and C. Schmid, "Aggregating Local Image Descriptors into Compact Codes.," *PAMI*, vol. 34, no. 9, 2012.

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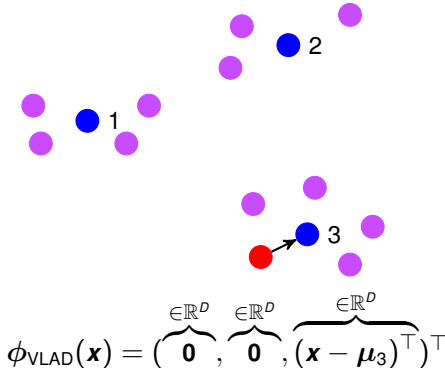


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Clusters: $\mathcal{D} = \{\boldsymbol{\mu}_k \in \mathbb{R}^D, k = 1, \dots, K\}$

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 Clusters: $\mathcal{D} = \{\boldsymbol{\mu}_k \in \mathbb{R}^D, k = 1, \dots, K\}$

$$\phi_k(\mathbf{x}) = \alpha_k(\mathbf{x})(\mathbf{x} - \boldsymbol{\mu}_k)$$

$$\alpha_k(\mathbf{x}) = \begin{cases} 1 & \text{if } k = \underset{j=1, \dots, K}{\operatorname{argmin}} \|\mathbf{x} - \boldsymbol{\mu}_j\|_2 \\ 0 & \text{else} \end{cases}$$

$$\phi_{\text{VLAD}}(\mathbf{x}) = (\phi_1^\top, \dots, \phi_K^\top)^\top \in \mathbb{R}^{D \cdot K}$$

⁴H. Jégou, F. Perronnin, M. Douze, J. Sánchez, P. Pérez, and C. Schmid, "Aggregating Local Image Descriptors into Compact Codes.," *PAMI*, vol. 34, no. 9, 2012.

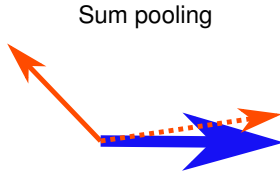


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Sum Pooling vs. Generalized Max Pooling

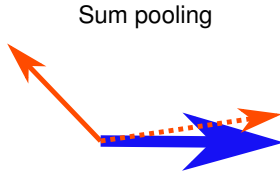


Visual Burstiness



- Unrelated descriptors produce interference
- Frequent descriptors dominate similarity

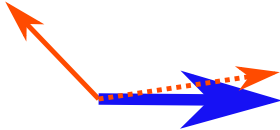
Visual Burstiness



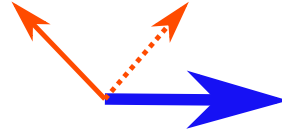
- Unrelated descriptors produce interference
- Frequent descriptors dominate similarity
- Choose better embedding
- Normalize encoding
 - Power normalization
 - Intra normalization
 - ...

Visual Burstiness

Sum pooling



Generalized max pooling [5]



- Unrelated descriptors produce interference
- Frequent descriptors dominate similarity
- Choose better embedding
- Normalize encoding
 - Power normalization
 - Intra normalization
 - ...

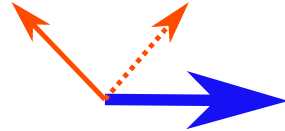
→ Balance pooling

Generalized Max Pooling

- Seek encoding ξ which weights each embedding ϕ

$$\xi = \sum_{\mathbf{x} \in \mathcal{X}} \alpha(\mathbf{x}) \phi(\mathbf{x}) = \alpha^T \Phi$$

Generalized max pooling [5]



Generalized Max Pooling

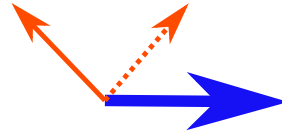
- Seek encoding ξ which weights each embedding ϕ

$$\xi = \sum_{\mathbf{x} \in \mathcal{X}} \alpha(\mathbf{x}) \phi(\mathbf{x}) = \alpha^T \Phi$$

- Max pooling: equally similar to frequent and rare patches
- Enforce similarity between any patch encoding and aggregated representation to be constant

$$\Phi^T \xi_{\text{gmp}} = \mathbf{1}_n,$$

Generalized max pooling [5]



Generalized Max Pooling

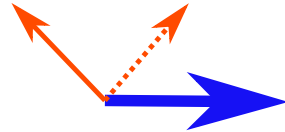
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$$\xi = \sum_{\mathbf{x} \in \mathcal{X}} \alpha(\mathbf{x}) \phi(\mathbf{x}) = \alpha^T \Phi$$

- Max pooling: equally similar to frequent and rare patches
- Enforce similarity between any patch encoding and aggregated representation to be constant

$$\Phi^T \xi_{\text{gmp}} = \mathbf{1}_n,$$

Generalized max pooling [5]



- Optimization problem can be cast as a ridge regression problem

$$\xi_{\text{gmp}} = \underset{\xi}{\operatorname{argmin}} \|\Phi^T \xi - \mathbf{1}_n\|^2 + \lambda \|\xi\|^2,$$

$\lambda \rightarrow 0$: max pooling

$\lambda \rightarrow \infty$: sum pooling



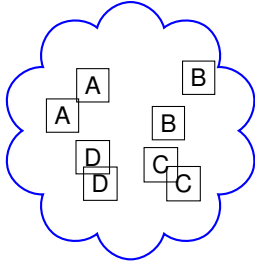
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Exemplar Classification

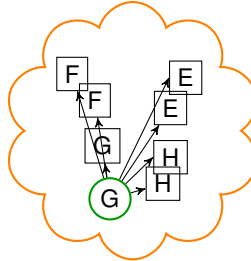


Similarity

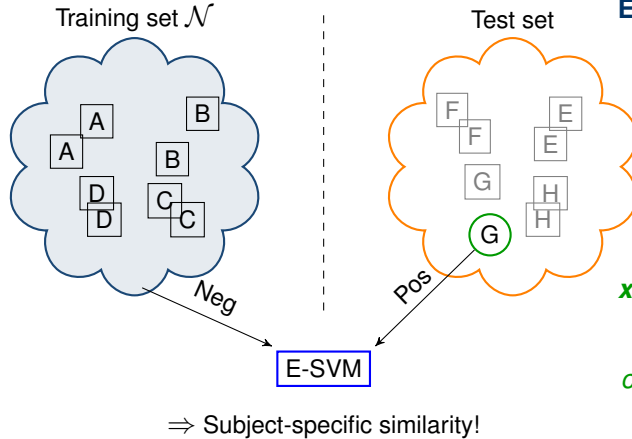
Training set $\mathcal{N}$



Test set



Similarity



Exemplar SVMs [Christlein17a]

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ & + c_p \max(0, 1 - \mathbf{w}^\top \mathbf{x}_p - b)^2 \\ & + c_n \sum_{\mathbf{x}_n \in \mathcal{N}} \max(0, 1 + \mathbf{w}^\top \mathbf{x}_n + b)^2 \end{aligned}$$

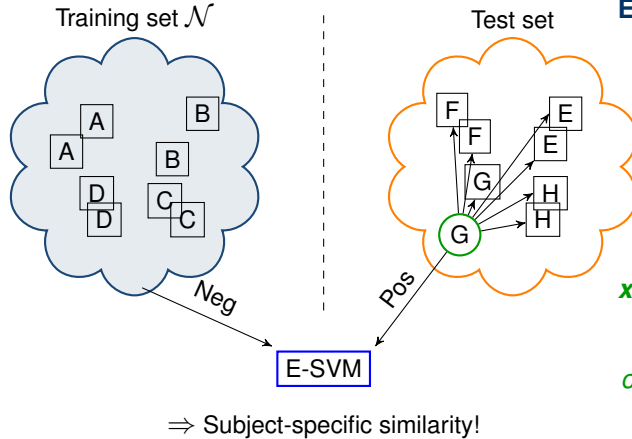
$\mathbf{x}_p, \mathbf{x}_n$: query sample, background sample

$\mathbf{w}, b$: model parameters

c_p, c_n : margin parameters

(e.g. indirect proportional to #samples)

Similarity



Exemplar SVMs [Christlein17a]

$$\begin{aligned} \min_{\mathbf{w}, b} & \frac{1}{2} \|\mathbf{w}\|^2 \\ & + c_p \max(0, 1 - \mathbf{w}^\top \mathbf{x}_p - b)^2 \\ & + c_n \sum_{\mathbf{x}_n \in \mathcal{N}} \max(0, 1 + \mathbf{w}^\top \mathbf{x}_n + b)^2 \end{aligned}$$

$\mathbf{x}_p, \mathbf{x}_n$: query sample, background sample

$\mathbf{w}, b$: model parameters

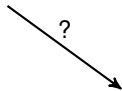
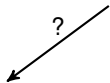
c_p, c_n : margin parameters

(e.g. indirect proportional to #samples)

Exemplar SVMs as Feature Encoder

- Trained SVM models: $\mathbf{w}$, b
 - Similarity independent of b when using cosine distance
- ⇒ New feature: $\mathbf{x} \rightarrow \frac{\mathbf{w}}{\|\mathbf{w}\|_2}$
- Note: iterative application of E-SVM possible but benefit vanishes quickly

Thank you for your attention



Questions? Questions? Questions?
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Questions? Questions? Questions?



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References



References I

- [1] F. Kleber, S. Fiel, M. Diem, and R. Sablatnig, “CVL-DataBase: An Off-Line Database for Writer Retrieval, Writer Identification and Word Spotting,” in *Document Analysis and Recognition (ICDAR), 2013 12th International Conference on*, Washington DC, NY, Aug. 2013, pp. 560–564.
- [2] G. Louloudis, B. Gatos, N. Stamatopoulos, and A. Papandreou, “ICDAR 2013 Competition on Writer Identification,” in *ICDAR*, Washington DC, NY, Aug. 2013, pp. 1397–1401.
- [3] S. Fiel, F. Kleber, M. Diem, V. Christlein, G. Louloudis, N. Stamatopoulos, and B. Gatos, “ICDAR2017 Competition on Historical Document Writer Identification,” in *ICDAR, 2013*.
- [4] D. G. Lowe, “Distinctive Image Features from Scale-Invariant Keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, Nov. 2004.
- [5] N. Murray, H. Jegou, F. Perronnin, and A. Zisserman, “Interferences in Match Kernels,” *TPAMI*, vol. 39, no. 9, 2016.
- [6] H. Jégou, F. Perronnin, M. Douze, J. Sánchez, P. Pérez, and C. Schmid, “Aggregating Local Image Descriptors into Compact Codes,” *PAMI*, vol. 34, no. 9, 2012.